Athul Anil Kumar

athul7744@outlook.com | linkedin.com/in/athul7744 | github.com/athul7744 | 91 801 55 70 147

About Me

Full-stack engineer skilled in **Java**, **Python**, and **JavaScript**, with a passion for **Machine Learning** and **AI**. I specialize in designing and implementing innovative solutions, such as contextual ML/AI features and Generative AI workflows, to enhance efficiency and customer satisfaction. Experienced in optimizing system performance, modernizing legacy frameworks, and delivering tailored solutions that drive adoption and streamline processes. I thrive on solving complex problems and am deeply curious about how things work. My goal is to use technology to create meaningful impact, making systems more efficient and leaving the world better than I found it.

Education

Georgia Institute of Technology — May 2025
Master of Science in Computer Science – Machine Learning — **GPA: 3.8/4.0**
Modules: Graduate Algorithms, Natural Language Processing, Quantum Computing, Deep Learning, Reinforcement Learning, Machine Learning, Game AI, Knowledge Based AI, HCI, Database Management

Sastra University — June 2019
Bachelor of Technology in Computer Science and Engineering (Honors) — **CGPA: 8.08/10.0**

Experience

Zoho Corporation | ServiceDesk Plus Cloud — June 2019 - Present
Member of Technical Staff — Chennai, India

- Led innovation in **ServiceDesk Plus** by implementing contextual ML/AI features, (RAG-based code generation, LLM-based summaries, similarity-based insights and translations), boosting customer satisfaction by **30%**
- Integrated **Agentic AI** across ServiceDesk Plus use cases, enabling seamless interoperability between products and streamlining cross-product workflows.
- Developed a declarative **feature development framework** using static definitions for Generative AI features, increasing developer productivity and reducing boilerplate code
- Spearheaded **Generative AI automations**, to reduce user onboarding effort and enhance platform capabilities
- Authored technical specs and use cases for **25+** features, aligning development with business goals and end-user needs
- Managed feature releases and migration of over **1M** records, with zero downtime and complete data integrity
- Collaborated with customers to gather feedback and deliver personalized solutions, driving a **35%** increase in adoption
- Optimized a legacy time-calculation engine, improving efficiency by **25%** through algorithmic enhancements.
- Boosted API performance by **20%** by offloading non-core functionality to an **Apache Kafka** queue
- Modernized backend systems by refactoring legacy Struts/JSP flows into a modular, **API-first** architecture (**Java 8, SQL, JS ES6**), improving **scalability** and **maintainability**.

Technical Skills

Skills: **Java, Python, Javascript, C#, SaaS, Full Stack Development, Machine Learning, NLP**
Frameworks/Tools: **Pytorch, Electron, Flutter, Apache Kafka, Flask, Git, Mercurial, NodeJS**
Databases: **MySQL, PostgreSQL**

Projects

Overcooked: Multi Agent Cooperative Reinforcement Learning

- Solving the overcooked multi agent environment with two cooks over several kitchen layouts using Deep Q Learning and Multi Agent Deep Deterministic Policy Gradient methods

Exploration of Image Captioning Methods

- Analyzed the effectiveness of image captioning methods from Convolutional and Recurrent Neural Networks to the latest Generative Transformers and compared their training and performance

Topic Adaptive Sentiment Analysis: Data Mining and Clustering

Submitted to ECML-PKDD

- Analyzing the initial sentiment of a person towards a topic and predicting how the sentiment changes after interacting with a polarizing topic

Certifications

An Introduction to Machine Learning — Andrew Ng at Stanford University
Bitcoin and Cryptocurrencies — BerkeleyX

Additional Information

Address: Flat No 2, Jaffar Residency, 9th Street, J Block, 15th Main Road, Anna Nagar, Chennai, Tamil Nadu, India
Languages: English, Tamil, Malayalam, Hindi
Date of Birth: 24 May 1997
Nationality: Indian